

S Aditha Krishna Bhat

Bengaluru, India | +91 98442 37346 | sadithakrishnabhat@gmail.com | linkedin.com/in/s-aditha-krishna-bhat

Summary

RTL Design Engineer with 5+ years of experience designing RISC-V SoC subsystems from microarchitecture definition through RTL implementation, CDC/LINT closure, and tape-out readiness. Experienced in translating architecture specifications into synthesizable RTL, driving design quality, and collaborating with architecture, verification, and firmware teams. Seeking opportunities in SoC Architecture and RTL Design where architectural exploration and implementation ownership are combined.

Experience

Hardware Engineer, ACL Digital – Bengaluru, India 2021 – Present

- **PROJECT 1 – RISC-V Input-Output Memory Management Unit (IoMMU):**

- Owned the microarchitecture definition and RTL implementation of an IoMMU block for a RISC-V SoC, translating architecture specifications into synthesizable SystemVerilog.
- Designed address translation, permission checking, and memory protection logic while ensuring seamless integration with the SoC memory subsystem.
- Collaborated with arch and verification teams throughout design reviews and RTL quality closure.

- **PROJECT 2 – RISC-V Advanced Interrupt Architecture (AIA):**

- Designed and implemented RTL modules for interrupt management compliant with the RISC-V Advanced Interrupt Architecture (AIA) specification.
- Participated in microarchitecture discussions to define block interfaces, CSR mapping, interrupt routing, and pipeline behavior.
- Performed CDC and LINT analysis using Synopsys tools, achieving clean sign-off, and developed register interfaces to accelerate firmware bring-up and system debugging.

- **PROJECT 3 – IP Verification & Design Quality Improvement:**

- Led design quality analysis for multiple communication interface IPs during SoC integration and tape-out preparation.
- Developed CDC constraint files, analyzed schematics and timing reports, and resolved integration issues identified through industry-standard EDA flows.
- Improved RTL quality by driving CDC/LINT closure and collaborating with multiple teams to deliver tape-out-ready IPs.

Education

IIT Roorkee, PG Certificate in VLSI Design Aug 2023

Bachelor of Engineering, Electrical and Electronics Engineering 2017 – 2021

- CGPA: 7.34 / 10

Technical Skills

Architecture & RTL Design: RISC-V Architecture, Microarchitecture, RTL Design, SystemVerilog, Verilog, FSM Design, SoC Integration

Design Verification & Quality: Code Coverage Analysis, CDC (Clock Domain Crossing), LINT, SpyGlass, JasperGold

EDA Tools: Synopsys Verdi, Synopsys Design Compiler, SpyGlass, Cadence Xcelium, QuestaSim, Questa Visualizer

Programming: SystemVerilog, Verilog, C, Python

Core Competencies: RTL Implementation, Architecture Specification Interpretation, Microarchitecture Development, CDC/LINT Closure, Design Quality Analysis, IP Integration, Cross-functional Collaboration